

Arjun Gupta

Embedded Systems & IoT Firmware Engineer • 6 Years Experience • MERIQ Index: 77% (Strong Candidate)

■ arjun.gupta425@example.com | ■ +1 (318) 714-6368 | ■ Chicago, IL

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, MQTT, I2C/SPI/UART, Python, FreeRTOS, Embedded Linux, PCB Debugging

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Tesla* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 44%.
- Led a squad of 5 engineers delivering scalable microservices processing 35M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 44%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2020 - 2023)

- Designed and maintained MQTT data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$23k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [C/C++, MQTT, I2C/SPI/UART, Python]

Engineered end-to-end production solution leveraging C/C++ and MQTT with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [I2C/SPI/UART, Python, FreeRTOS, Embedded Linux]

Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Computer Science

University of Toronto — 2016 - 2020 (Score: 3.44 / 4.0)